Rock strata

Sedimentary rocks were laid down horizontally to form layers called strata, with the oldest at the bottom. This is the basis of the relative dating system called the geological timescale. But finding the precise age of rock strata was impossible until recently, when scientists found out how to detect radioactive elements. These have enabled scientists to determine not only the age of rocks, but also the age of planet Earth itself.

ROCK RECORD

Wherever rocks are forming today—usually deep beneath lake beds, or on the bottom of the sea—they are being created from soft sediments that settle in horizontal, parallel layers. A new layer may be very similar to the one that it settles on, or it may be visibly different due to some change in the environment. A huge volcanic eruption, for example, might eject ash that settles in a distinctive layer. The oldest layers are at the bottom.

BEND AND TILT

The moving plates of Earth's crust can bend and fold rock strata so they are no longer horizontal. Often, they seem to be tilted, or even vertical, because the folds in the layers are so big that we can see only parts of them. But sometimes the rock strata are so crumpled up by extreme pressure that they form dramatic zigzag patterns.